

Manufacturing Execution System Analysis Report

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Executive Summary

Software update on Jul 3 2026 triggered control system error causing 30 scrap units peak 22 on Jul 3 with financial impact of \$7600 to \$15200. Root cause identified and resolved by Jul 5 corrective fix.

Title

Battery Cell Variance Defect Analysis Report

Period

July 1-7, 2026

Scope

Maintenance Correlation

Root Causes

- {'rank': 1, 'confidence': '95%', 'description': 'Software-induced control system failure on Bat-40 Bat-41 Qua-70 Qua-71'}
- {'rank': 2, 'confidence': '88%', 'description': 'No software change management validation gate before production deployment'}
- {'rank': 3, 'confidence': '82%', 'description': 'Calibration degradation 233% increase in maintenance duration'}
- {'rank': 4, 'confidence': '70%', 'description': '2-hour detection lag 8 work orders processed before detection'}

Action Plan

Immediate: ['Enforce software change management protocol', 'Deploy real-time maintenance duration monitoring']

Short Term: ['Implement SPC triggers', 'Complete calibration audit Bat-40 Bat-41 Qua-70 Qua-71']

Long Term: ['Investigate OEE 0.33-0.39 on Battery Assembly', 'Full MES software governance framework']

Kpis

- Zero unvalidated software deployments
- Maintenance alert response under 5 min
- SPC halt within 2 units of deviation
- OEE improvement above 0.75

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